

Back to single-variable Calculus

Arc-Length

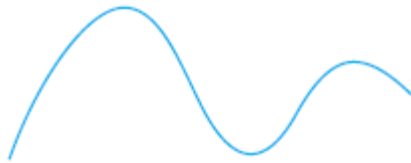


FIGURE 1

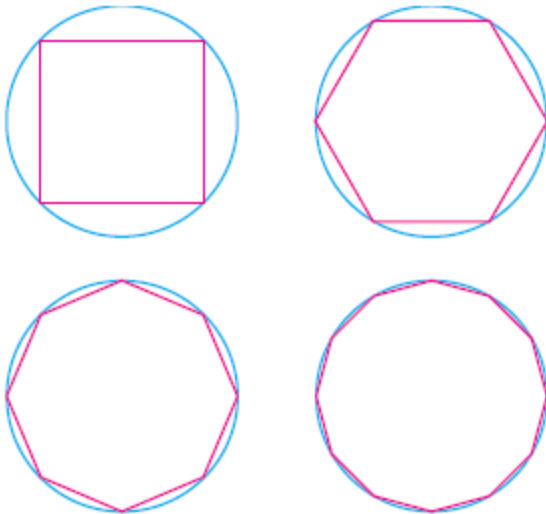
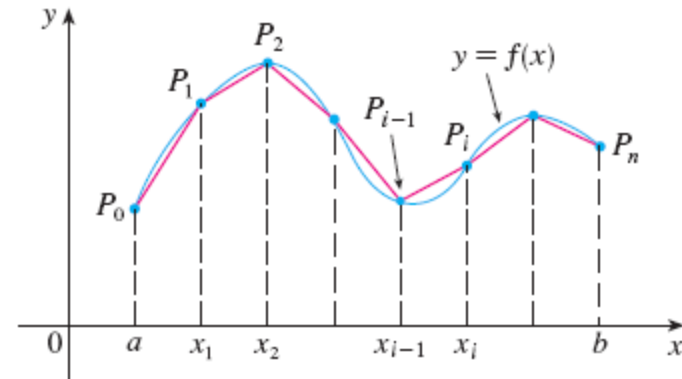


FIGURE 2



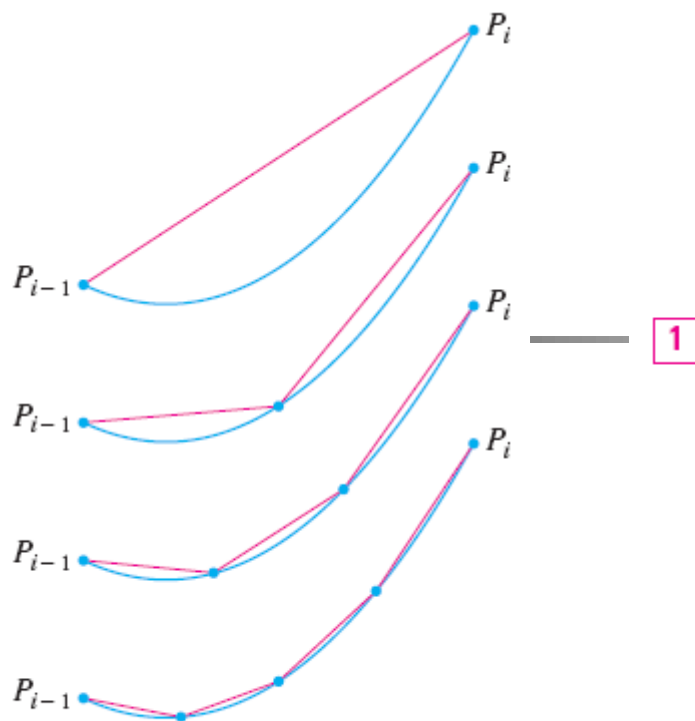


FIGURE 4

$$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n |P_{i-1}P_i|$$

2 The Arc Length Formula If f' is continuous on $[a, b]$, then the length of the curve $y = f(x)$, $a \leq x \leq b$, is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} \, dx$$

3

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

If a curve has the equation $x = g(y)$, $c \leq y \leq d$, and $g'(y)$ is continuous, then by interchanging the roles of x and y in Formula 2 or Equation 3, we obtain the following formula for its length:

4

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} \, dy = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy$$

The Arc Length Function

We will find it useful to have a function that measures the arc length of a curve from a particular starting point to any other point on the curve. Thus if a smooth curve C has the equation $y = f(x)$, $a \leq x \leq b$, let $s(x)$ be the distance along C from the initial point $P_0(a, f(a))$ to the point $Q(x, f(x))$. Then s is a function, called the **arc length function**, and, by Formula 2,

5

$$s(x) = \int_a^x \sqrt{1 + [f'(t)]^2} \, dt$$

6

$$\frac{ds}{dx} = \sqrt{1 + [f'(x)]^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

Equation 6 shows that the rate of change of s with respect to x is always at least 1 and is equal to 1 when $f'(x)$, the slope of the curve, is 0. The differential of arc length is

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

and this equation is sometimes written in the symmetric form

$$(ds)^2 = (dx)^2 + (dy)^2$$

Geometric Interpretation:

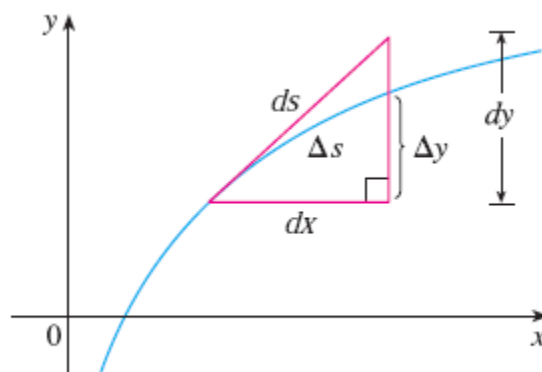


FIGURE 7

Area of Surface of Revolution:

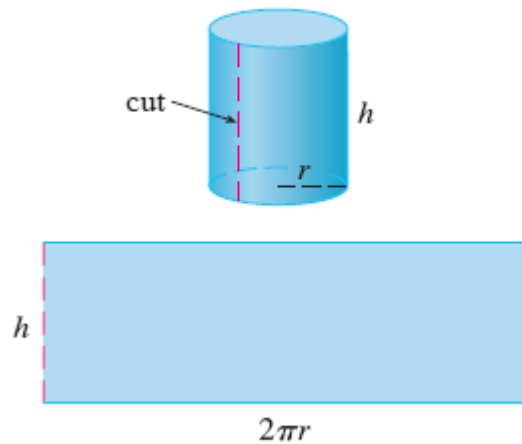
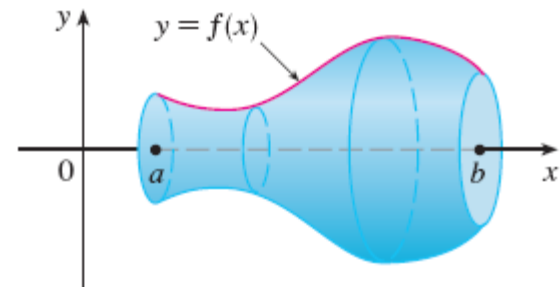
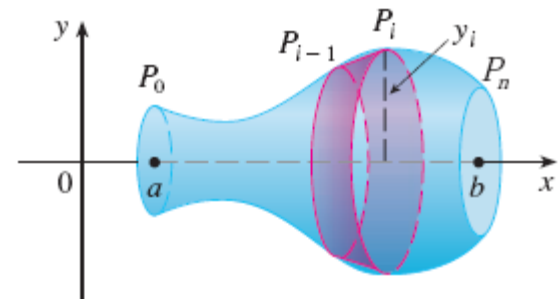


FIGURE 1



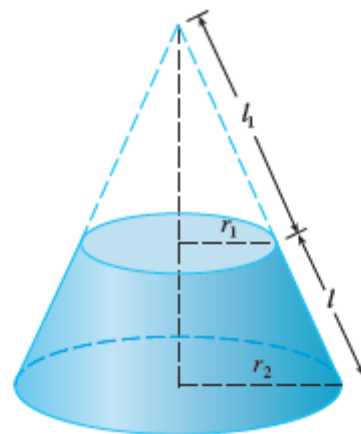
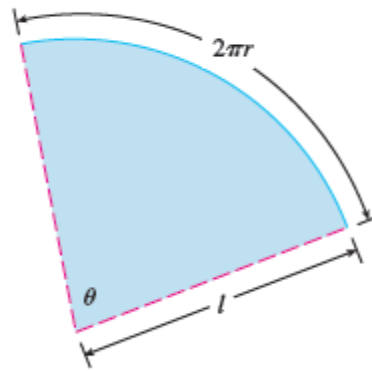
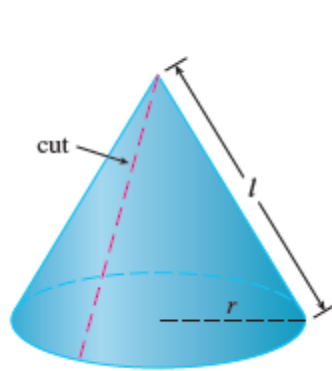
(a) Surface of revolution



(b) Approximating band

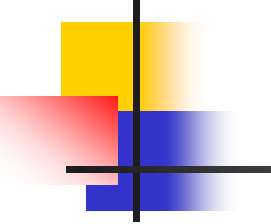
FIGURE 4

The lateral surface area of a cone to be $A = \pi rl$.



$$A = \pi(r_1 l + r_2 l)$$

$$A = 2\pi rl$$



Therefore, in the case where f is positive and has a continuous derivative, we define the **surface area** of the surface obtained by rotating the curve $y = f(x)$, $a \leq x \leq b$, about the x -axis as

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$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx$$

With the Leibniz notation for derivatives, this formula becomes

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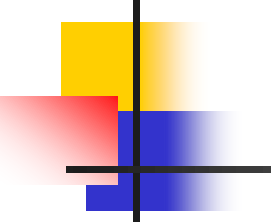
$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

If the curve is described as $x = g(y)$, $c \leq y \leq d$, then the formula for surface area becomes

6

$$S = \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

and both Formulas 5 and 6 can be summarized symbolically, using the notation for arc length given in Section 8.1, as



7

$$S = \int 2\pi y \, ds$$

For rotation about the y -axis, the surface area formula becomes

8

$$S = \int 2\pi x \, ds$$

V EXAMPLE 1 The curve $y = \sqrt{4 - x^2}$, $-1 \leq x \leq 1$, is an arc of the circle $x^2 + y^2 = 4$. Find the area of the surface obtained by rotating this arc about the x -axis. (The surface is a portion of a sphere of radius 2. See Figure 6.)

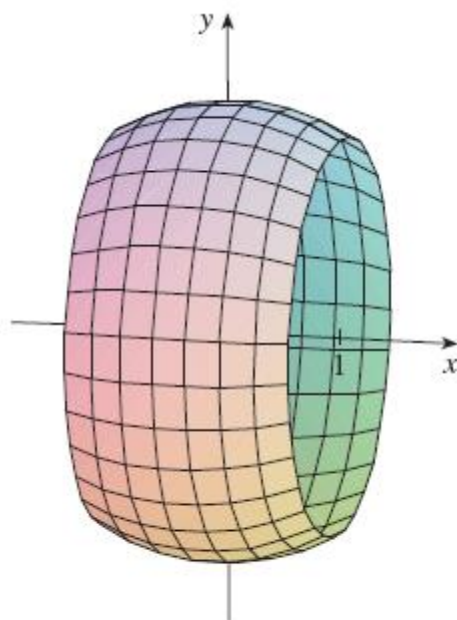


FIGURE 6

V EXAMPLE 2 The arc of the parabola $y = x^2$ from $(1, 1)$ to $(2, 4)$ is rotated about the y -axis. Find the area of the resulting surface.

